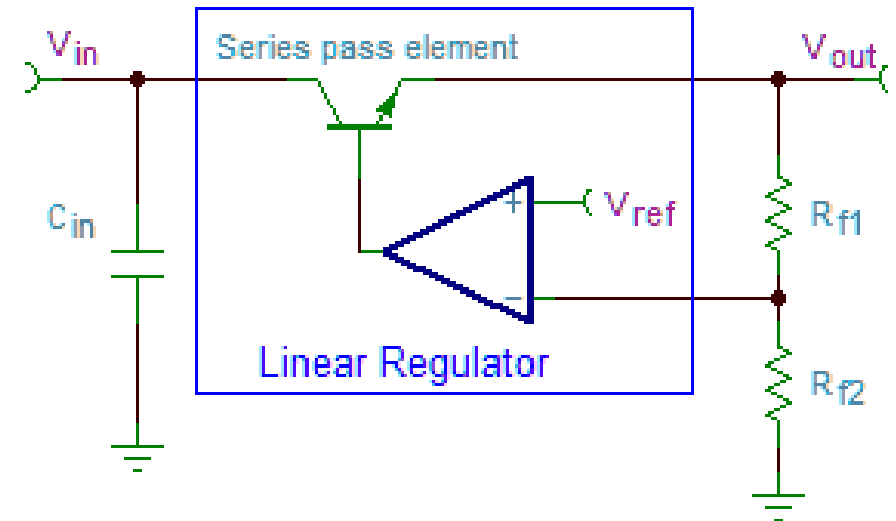


DC-DC CONVERTERS – BUCK CONVERTERS

BUCK1-1 LINEAR vs SWITCHING

It was mentioned at the beginning of this course that it was possible to do power conversion with either a linear system, or with a switching system.

Linear voltage conversion usually involves dissipating controlled amounts of power from effectively resistive elements to provide a desired output voltage. Depending on the amount of power purposefully dissipated, this type of voltage conversion can be electrically very inefficient.



Switching voltage conversion on the other hand controls output voltage through limiting input power transfer by switching. Unwanted fluctuations in output voltage caused by the switching are handled by use of appropriate filtering (low loss). Since appreciable power is not dissipated or absorbed, switching voltage conversion can be very efficient. As we will soon see, the switch/filter combination can also provide extra voltage conversion functionality not easily achieved with linear systems.

BUCK1-1.1 Basic Switching Converter Concept

Figure 1.1 shows a basic switching converter and its associated output waveform (for a purely resistive load). Obviously, **on average** over the period T the load resistor voltage is less than the source DC voltage.

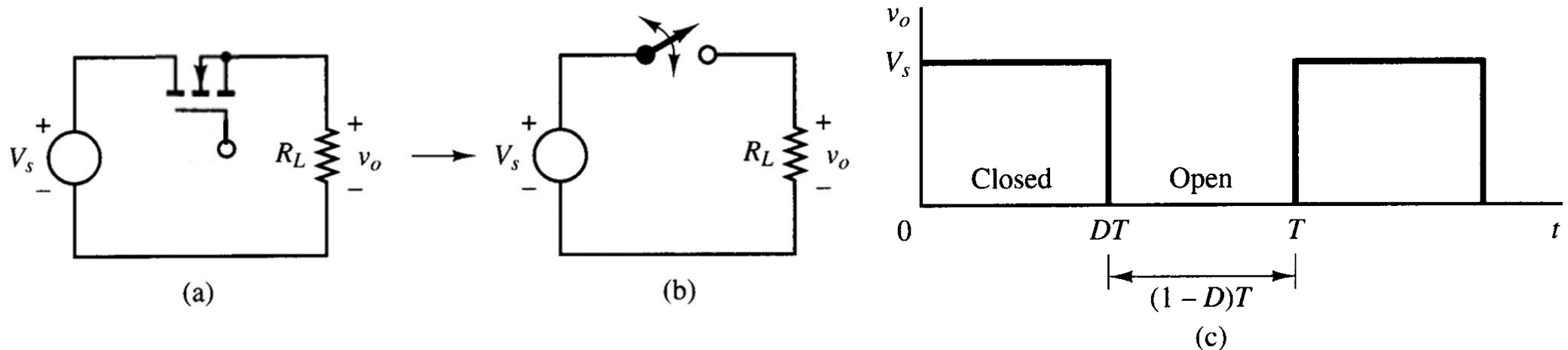


Figure 1.1 Basic DC-DC switching converter showing load voltage. D is the duty ratio (or duty cycle).

The concept of **duty ratio D** (also commonly known as duty cycle) was introduced in the last lecture. Duty ratio is a measure of the proportion of an entire period that the switch is **on**, such that,

$$D \equiv \frac{t_{on}}{T} = t_{on} f$$

So in this case, the average load voltage over one period is simply DV_s . Changing the time width of the DC voltage by changing D , thus changing the output voltage is one example of what is known as **Pulse Width Modulation (PWM)** – the output value is modulated by the pulse width.

BUCK1-2 THE BUCK CONVERTER

Most applications of DC power usually require a fairly steady DC voltage (small voltage ripple). The basic converter shown in Section BUCK1-1.1 has large fluctuations (from 0V to $+V_S$), so is not particularly useful as is. What's needed is an output filter that incorporates energy recovery. That's what a **buck converter** is - a basic step-down converter with an energy recovery filter added on. The buck converter topology and essential operation is shown in Figure 2.1.

One of the assumptions we will make in the analysis of the DC-DC converters covered in this course is that the **inductor current is continuous** (unless stated otherwise) such that, after first turning on, $i_L(t)$ never falls to zero. This condition is not essential for converter operation, but is usually designed for and is certainly easier to analyse.

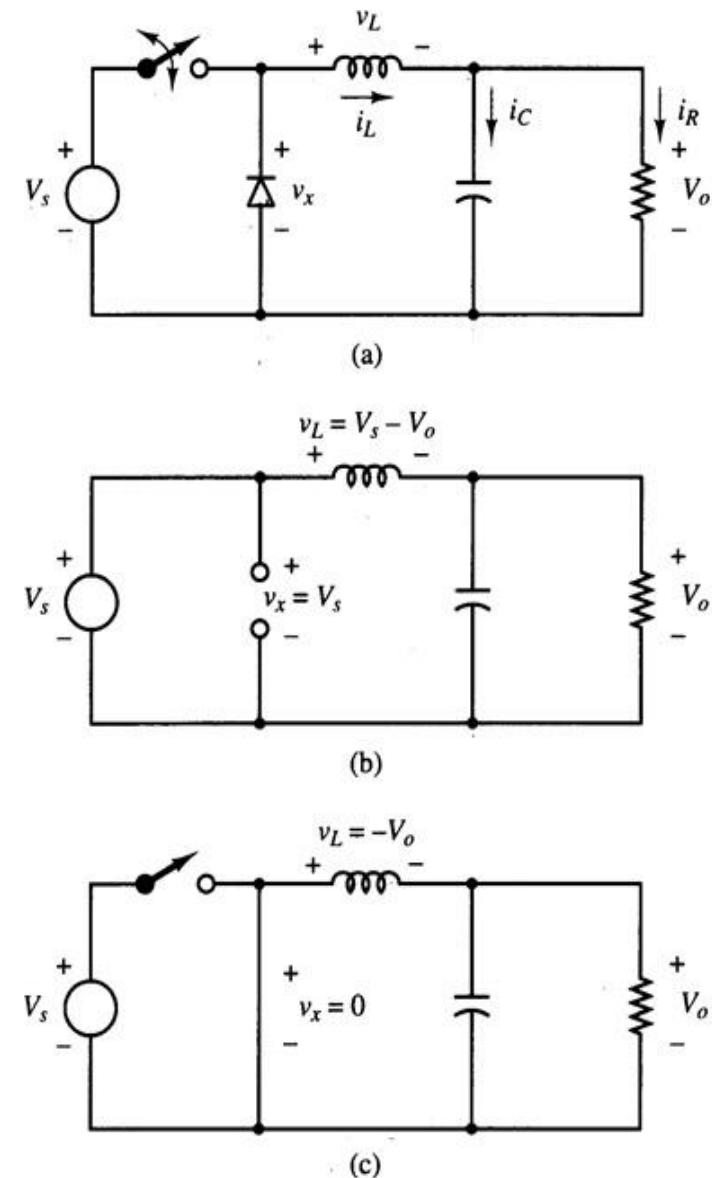


Figure 2.1 Buck converter showing operation when switch is on and off.

Also, in general, the DC-DC converters we will be looking at **all** have the following **steady state** properties;

- The inductor current fluctuations are periodic $\{i_L(t+T) = i_L(t)\}$.
- The average “change” in inductor current is zero.
- The average inductor voltage is zero.
- The average capacitor current is zero.
- The power supplied by the source equals the power delivered to the load (plus any non-ideal losses).

We will also be making the simplifications in the analysis that the components used are ideal.

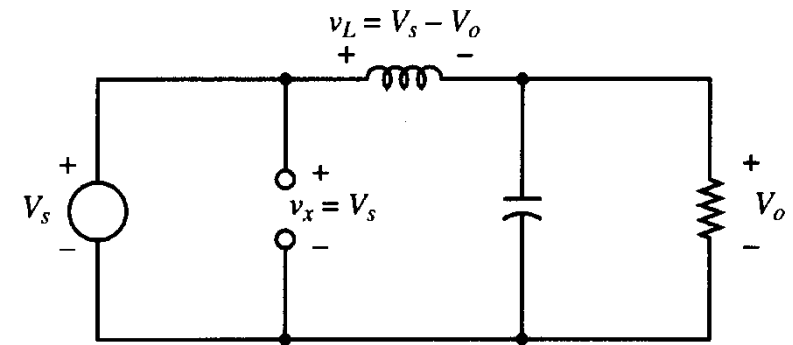
BUCK1-2.1 Buck Converter Analysis

Since it is the inductor current that ultimately determines output voltage in the DC-DC converters we look at, the key analysis looks at that inductor current.

Switch Closed

When the switch is closed, the diode is reverse biased and the inductor voltage is just,

$$v_L = L \frac{di_L}{dt} = V_S - V_O$$



Since V_S **and** V_O are constant (remember steady state assumptions), then the time derivative of the inductor current is a positive constant. That means, the inductor current must be **constantly ramping up** while the switch is closed, where

$$\Delta i_{L, closed} = \left(\frac{V_S - V_O}{L} \right) DT$$

Switch Open

When the switch is open, the inductor voltage reverses to keep the current flowing in the same direction, which forward biases the diode. In this situation the inductor is in parallel with the load.

So,

$$v_L = L \frac{di_L}{dt} = -V_o$$

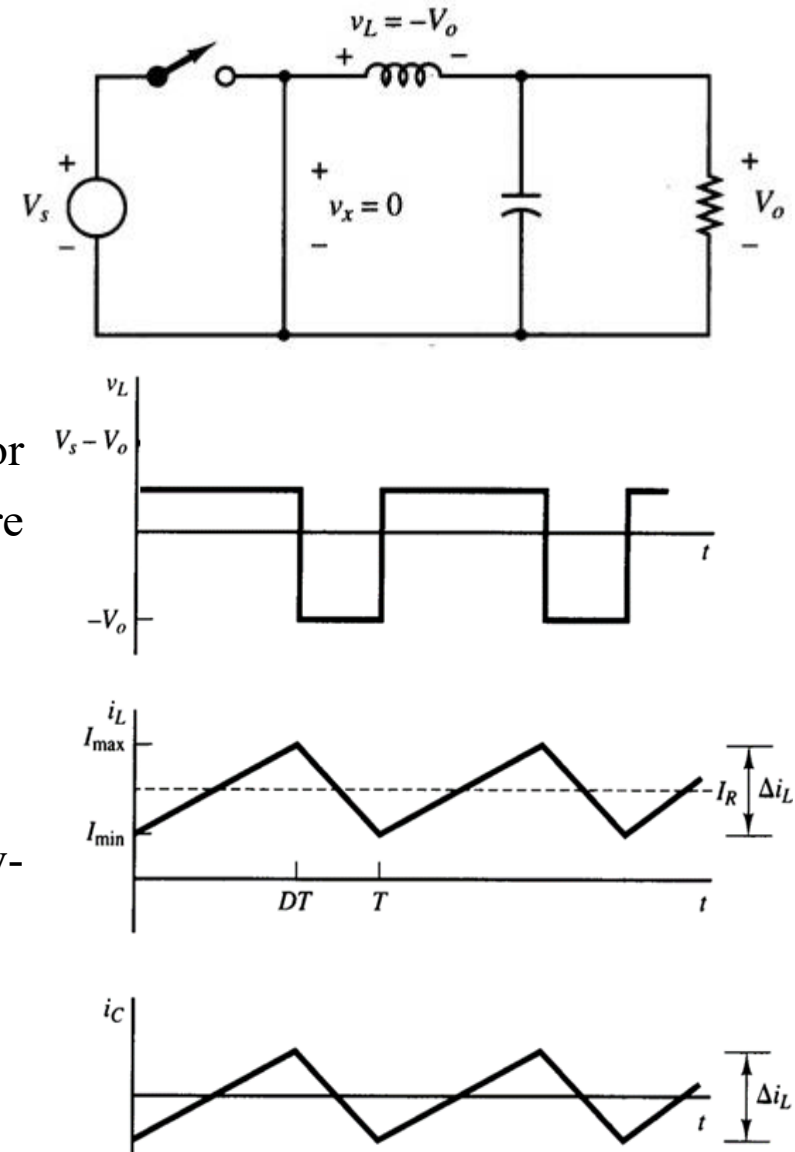
Again the derivative of the current is a constant, so the inductor current is **constantly ramping down** while the switch is open, where

$$\Delta i_{L,open} = -\left(\frac{V_o}{L}\right)(1-D)T$$

Steady State Result

The average fluctuation in inductor current must equal zero (steady-state). As such, we can state,

$$\Delta i_{L,closed} + \Delta i_{L,open} = 0$$



Here's something to consider. What would make the inductor current of the buck converter become discontinuous?

Source current?